



JCB Migrates 80 Systems to AWS, Advancing Common Infrastructure and Data Infrastructure to Respond to Changes in the Business Environment

Learn how JCB in finance migrated several business systems to AWS, saving IT costs and improving business agility.

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80 systems

scheduled for migration to AWS

100+ systems

linked using an API

30%

expected cost savings



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Overview

One of the world's seven major credit card brands, [JCB](#) is the only brand originating in Japan. To adapt to changes in the business environment, the company is migrating business systems that have been operating on premises since 2017 to Amazon Web Services (AWS). In addition to anticipating a 30 percent reduction in operation and other infrastructure costs through standardization, the company has begun implementing business initiatives based on common infrastructure and data infrastructure.



Opportunity | Embracing Cloud Infrastructure to Provide Engaging Services

JCB is the only international credit card company based in Japan, and it provides a variety of financial services. Its annual transaction volume is more than 37.7 trillion yen (based on data from the fiscal year of 2021), and the number of JCB cardholders reached 150.56 million by the end of September 2022.

"In recent years, the use of modern financial services like electronic money and QR code payments has increased, and competition for financial services has intensified," says Ichiro Nakata, executive officer and head of system headquarters at JCB. "As a result of the increase in frequency of small payment amounts due to the spread of cashless payments, revenue obtained from a single transaction is also on a downward trend. To adapt to these changes and to continue to provide engaging services, the three key management issues are improving business agility, facilitating business continuity, and optimizing IT costs."



Therefore, the company set out to migrate its on-premises business systems to the cloud and launched a system infrastructure optimization project in the fiscal year of 2017.

Solution | Building an API Infrastructure and a Data Lake

At the 2018 management meeting, JCB unanimously decided to use AWS.

The decisive factor in choosing AWS was our track record at financial institutions such as major banks,” says Nakata. “Another point is that in terms of security required of financial institutions, it is possible to comply with Payment Card Industry Data Security Standard (PCI DSS) certification and The Center for Financial Industry Information Systems (FISC) safety standards using AWS.”

“By using AWS PCI DSS certification, the burden of obtaining PCI DSS certification for the entire system has been reduced,” says Keisuke Hisaoka from the digital solution development department of the system headquarters at JCB.

In migrating existing systems, the company decided to adopt a “shift” that maximized the benefits unique to the cloud rather than a “lift” that would directly migrate the existing configuration to the cloud. JCB formed a virtual organization across system headquarters to use AWS. Referring to the [AWS Cloud Adoption Framework](#) (AWS CAF), the company adopted a six-session method and promoted policies centered on business, human resources, governance, operations, solutions, and security.

To migrate with an effective shift, JCB implemented various support measures for each system development process. At the examination stage, each system was mapped in four quadrants centering on migration difficulty and migration benefits. This way, JCB could consider the balance between the difficulty level and the benefits to decide which systems were desirable to migrate, and in what form, so that the system management department could evaluate the direction of the migration in advance.

The company gradually migrated to 2 systems in the fiscal year of 2018 and expanded to 11 systems in the fiscal year of 2019. As of March 2023, JCB has 54 systems running on AWS. Going forward, the company plans to migrate



half of the 160 systems managed by the system headquarters by the end of the fiscal year of 2024.

In parallel with the system migration, JCB moved forward with the construction of a common infrastructure and the construction of a data lake.

One part of the common infrastructure is the API infrastructure, which relays between external financial institutions and settlement companies and internal business systems. “Until now, we had integrated individual systems, but doing so is time consuming and costly,” says Kyoko Kashimata from the digital solution development department of the system headquarters at JCB. “Therefore, we decided to build an API solution on AWS to efficiently exchange data.”

The API infrastructure consists of a scalable architecture that uses managed services. These include database services like [Amazon Aurora](#), [Amazon DynamoDB](#). Additionally, the API infrastructure is protected using [AWS WAF](#). “By using AWS, it is now possible to simply upgrade middleware after operation, increase memory, and scale with the increase in traffic,” says Takada Yusuke from the digital solution development department of the system headquarters at JCB.

It usually takes 6 months to procure and build infrastructure, but building an API infrastructure has been reduced to 3 months for JCB. Since their release in July 2019, the number of linked systems using APIs has continued to grow, and the number of APIs, which was in the single digits, increased to more than 100 at the end of 2022.

The data infrastructure is for centrally managing internal and external data and providing an analysis environment for all employees. “In the on-premises data warehouses, it was difficult to expand storage, and most of the data to be analyzed was limited to a storage period of 2 years or less,” says Taiki Takekuni from the business system development department of the system headquarters at JCB. “The current data infrastructure targets all kinds of data, and it was built on [Amazon Simple Storage Service](#) (Amazon S3), which is inexpensive and scalable, to analyze data over a long period of 5–10 years.” Using Amazon S3, the company seamlessly implements collaboration between tools for data analysis.

Using AWS, JCB has an environment that facilitates fast analysis. The company processes data for analysis using [Amazon Athena](#) and catalogs information for data accumulated using [AWS Glue](#).

“As of October 2022, approximately 32 TB of data have been stored in the data infrastructure,” says Takekuni. “We plan to expand the data to around 120 TB by 2027.”

In the future, in order to further expand the amount and type of third-party and other data and to provide data in a form that is simple to use, the data infrastructure is to promote cooperation with other analysis solutions using [AWS PrivateLink](#), without exposing data to the internet.

“Currently, we’re using tools on the data infrastructure to anonymize transaction data for JCB Consumption NOW, which is a dashboard about the JCB credit card holders’ transactions,” says Takekuni. “In the future, we will revitalize data use while cooperating with AI and data analysis solutions operated by JCB and running on AWS.”

Architecture

AWS Services Used

AWS Cloud Adoption Framework (AWS CAF)

The AWS Cloud Adoption Framework (AWS CAF) leverages AWS experience and best practices to help you digitally transform and accelerate your business outcomes through innovative use of AWS.

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Amazon Aurora

Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication, and integrations with other AWS services.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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Amazon Athena

Amazon Athena is a serverless, interactive analytics service built on open-source frameworks, supporting open-table and file formats.

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